

Sector	Domain	Primary Problem Solved	Key Technologies	Typical Roles	Leading Companies	Future Growth
Artificial Intelligence	Generative AI	Building intelligent assistants and automation systems	Python PyTorch TensorFlow Transformers LLMs	ML Engineer AI Engineer Research Engineer	OpenAI, Google, Microsoft, Anthropic, Meta	Very High
Artificial Intelligence	Machine Learning	Prediction classification and decision automation	Python Statistics Scikit-learn PyTorch ML Algorithms	ML Engineer Data Scientist	Google, NVIDIA, Amazon, Meta	Very High
Artificial Intelligence	Computer Vision	Understanding images videos and physical environments	CNNs Vision Transformers OpenCV CUDA	Computer Vision Engineer	NVIDIA, Tesla, Google, Apple	Very High
Artificial Intelligence	Natural Language Processing	Understanding and generating human language	NLP Transformers BERT LLMs Python	NLP Engineer	Google, OpenAI, Microsoft, Meta	Very High
Cloud Computing	Cloud Infrastructure	Running scalable applications globally	AWS Azure GCP Kubernetes Docker Terraform	Cloud Engineer DevOps Engineer	Amazon, Microsoft, Google	Very High
Cloud Computing	Distributed Systems	Building reliable large-scale systems	System Design Networking Databases Go Java	CLOUD Engineer Backend Engineer	Google, Amazon, Netflix, Uber	Very High
Cloud Computing	Platform Engineering	Internal developer platforms and automation	Kubernetes CI/CD Containers Go Python	Platform Engineer SRE	Spotify, Google, Microsoft	High
Software Engineering	Backend Development	Building APIs and business logic	Java Go Python C++ SQL REST GraphQL	Backend Engineer	Amazon, Google, Meta, Stripe	Very High
Software Engineering	Frontend Development	Building user interfaces and experiences	JavaScript TypeScript React Angular CSS	Frontend Engineer	Google, Microsoft, Airbnb	High
Software Engineering	Full Stack Development	Complete application development	Frontend Backend Databases Cloud	Full Stack Engineer	Startups, SaaS Companies, Big Tech	High
Data Engineering	Data Platforms	Collecting storing and processing massive data	SQL Python Spark Kafka Airflow	Data Engineer	Databricks, Netflix, Amazon	Very High
Data Science	Analytics and Insights	Using data for business decisions	Python SQL Statistics Visualization	Data Analyst Data Scientist	Uber, Airbnb, Microsoft	High
Database Engineering	Database Systems	Managing high-scale data storage	SQL NoSQL Distributed Databases C++	Database Engineer	Google, Amazon, Oracle	High
Cybersecurity	Application Security	Protecting software from attacks	Security Testing Encryption OWASP	Security Engineer	Microsoft, CrowdStrike, Palo Alto Networks	Very High
Cybersecurity	Cloud Security	Protecting cloud infrastructure	IAM Security Tools Networking	Cloud Security Engineer	AWS, Google, Microsoft	Very High
FinTech	Digital Payments	Moving money securely worldwide	APIs Cryptography Banking Systems	Backend Engineer FinTech Engineer	Stripe, Visa, Mastercard, PayPal	Very High
FinTech	Quantitative Finance	Algorithmic trading and financial modeling	C++ Python Mathematics Statistics	Quant Developer	Jane Street, Goldman Sachs, Citadel	High
Enterprise Software	SaaS Platforms	Business workflow automation	Cloud APIs Databases Microservices	SaaS Engineer	Salesforce, ServiceNow, Atlassian	High
Enterprise Software	ERP Systems	Enterprise resource planning	Business Logic Databases Cloud	Enterprise Engineer	SAP, Oracle, Microsoft	High
Semiconductors	AI Hardware	Building chips for AI computation	VLSI CUDA Hardware Design C++	Hardware Engineer	NVIDIA, AMD, Intel	Very High
Semiconductors	Processor Design	Creating CPUs and computing hardware	Computer Architecture Verilog C++	Chip Design Engineer	Apple, Intel, Qualcomm	High
Robotics	Industrial Robotics	Automating manufacturing processes	C++ ROS Computer Vision Control Systems	Robotics Engineer	ABB, Tesla, FANUC	Very High
Autonomous Vehicles	Self Driving Technology	Automated transportation systems	CV AI Sensors Robotics C++	Autonomous Engineer	Tesla, Waymo, NVIDIA	Very High
Aerospace	Space Technology	Rockets satellites and space systems	Physics Embedded Systems AI	R&D Engineer	SpaceX, Boeing, Lockheed Martin	High
Healthcare Technology	Medical AI	Improving diagnosis and treatment	AI Imaging Data Science	Healthcare AI Engineer	Google Health, Microsoft, NVIDIA	Very High
Biotechnology	Drug Discovery	Creating medicines faster	AI Biology Data Science	Bioinformatics Engineer	Moderna, Pfizer, Roche	High
E-Commerce	Online Marketplaces	Connecting buyers and sellers	Recommendation Systems Payments Cloud	Software Engineer	Amazon, Alibaba, Shopify	High
Recommendation Systems	Personalization	Improving user experiences	Machine Learning Data Engineering	ML Engineer	Netflix, Spotify, YouTube	Very High
Networking	Internet Infrastructure	Connecting billions of devices	TCP/IP Routing Network Hardware	Network Engineer	Cisco, Cloudflare, Google	High
Telecommunications	5G and Wireless	High-speed communication networks	RF Engineering Networking Embedded	Wireless Engineer	Qualcomm, Ericsson, Nokia	High
Energy Technology	Renewable Energy	Reducing carbon emissions	Batteries AI Smart Grid Systems Engineer	Tesla, Siemens, GE	High	
Manufacturing	Industry 4.0	Smart factories and automation	IoT Robotics AI Automation Engineer	Siemens, Bosch, Honeywell	High	
Logistics	Supply Chain Technology	Optimizing global movement of goods	AI Optimization Cloud Data	Logistics Engineer	Amazon, DHL, FedEx	High
Gaming	Game Technology	Interactive entertainment experiences	C++ Graphics Engines AI	Game Engineer	Microsoft Xbox, Sony, Epic Games	High
AR/VR	Spatial Computing	Digital interaction with environments	Computer Vision 3D Graphics AI	XR Engineer	Apple, Meta, Microsoft	Very High